
14

Inductor Design

This chapter treats the design of magnetic elements such as filter inductors, using the K_g method. With this method, the maximum flux density B_{max} is specified in advance, and the element is designed to attain a given copper loss.

The design of a basic filter inductor is discussed in Sections 14.1 and 14.1.5. In the filter inductor application, it is necessary to obtain the required inductance, avoid saturation, and obtain an acceptable low dc winding resistance and copper loss. The geometrical constant K_g is a measure of the effective magnetic size of a core, when dc copper loss and winding resistance are the dominant constraints [1,2]. Design of a filter inductor involves selection of a core having a K_g sufficiently large for the application, then computing the required air gap, turns, and wire size. A simple step-by-step filter inductor design procedure is given. Values of K_g for common ferrite core shapes are tabulated in Appendix D.

Extension of the K_g method to multiple-winding elements is covered in Section 14.3. In applications requiring multiple windings, it is necessary to optimize the wire sizes of the windings so that the overall copper loss is minimized. It is also necessary to write an equation that relates the peak flux density to the applied waveforms or to the desired winding inductance. Again, a simple step-by-step transformer design approach is given.

The goal of the K_g approach of this chapter is the design of a magnetic device having a given copper loss. Core loss is not specifically addressed in the K_g approach, and B_{max} is a given fixed value. In the next chapter, the flux density is treated as a design variable to be optimized. This allows the overall loss (i.e., core loss plus copper loss) to be minimized.

14.1 FILTER INDUCTOR DESIGN CONSTRAINTS

A filter inductor employed in a CCM buck converter is illustrated in Fig. 14.1(a). In this application, the value of inductance L is usually chosen such that the inductor current ripple peak magnitude Δi is a small fraction of the full-load inductor current dc component I , as illustrated in Fig. 14.1(b). As illustrated in Fig. 14.2, an air gap is employed that is sufficiently large to prevent saturation of the core by the peak

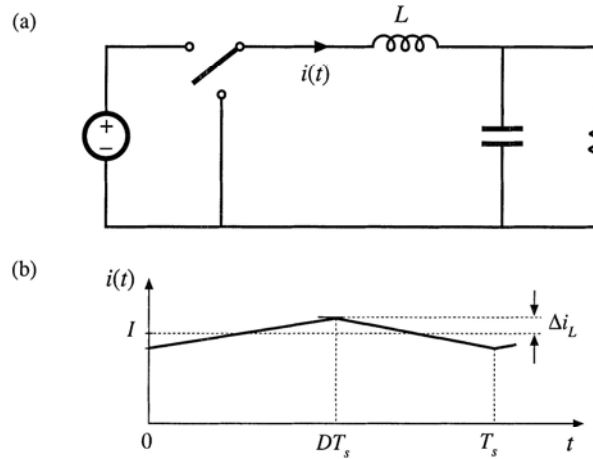


Fig. 14.1 Filter inductor employed in a CCM buck converter: (a) circuit schematic, (b) inductor current waveform.

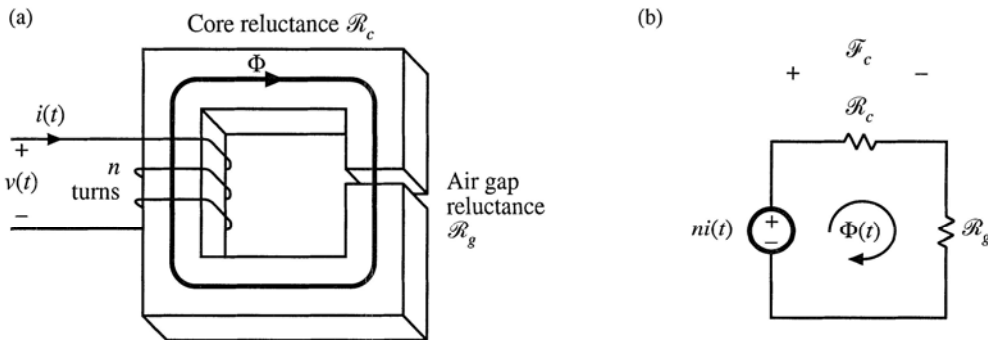


Fig. 14.2 Filter inductor: (a) structure, (b) magnetic circuit model.

current $I + \Delta i$.

Let us consider the design of the filter inductor illustrated in Figs. 14.1 and 14.2. It is assumed that the core and proximity losses are negligible, so that the inductor losses are dominated by the low-frequency copper losses. The inductor can therefore be modeled by the equivalent circuit of Fig. 14.3, in which R represents the dc resistance of the winding. It is desired to obtain a given inductance L and given winding resistance R . The inductor should not saturate when a given worst-case peak current I_{max} is applied. Note that specification of R is equivalent to specification of the copper loss P_{cu} , since

$$P_{cu} = I_{rms}^2 R \quad (14.1)$$

The influence of inductor winding resistance on converter efficiency and output voltage is modeled in Chapter 3.

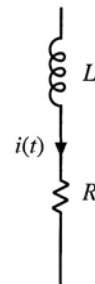


Fig. 14.3 Filter inductor equivalent circuit.

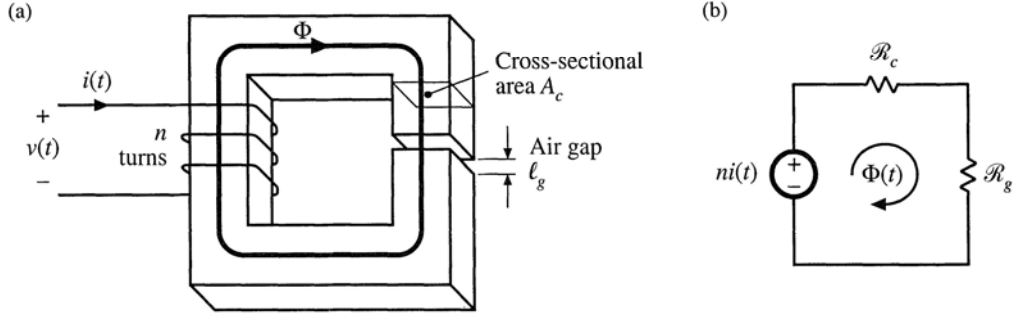


Fig. 14.4 Filter inductor: (a) assumed geometry, (b) magnetic circuit.

It is assumed that the inductor geometry is topologically equivalent to Fig. 14.4(a). An equivalent magnetic circuit is illustrated in Fig. 14.4(b). The core reluctance $\mathcal{R}_c$ and air gap reluctance $\mathcal{R}_g$ are

$$\begin{aligned}\mathcal{R}_c &= \frac{\ell_c}{\mu_c A_c} \\ \mathcal{R}_g &= \frac{\ell_g}{\mu_0 A_c}\end{aligned}\quad (14.2)$$

where ℓ_c is the core magnetic path length, A_c is the core cross-sectional area, μ_c is the core permeability, and ℓ_g is the air gap length. It is assumed that the core and air gap have the same cross-sectional areas. Solution of Fig. 14.4(b) yields

$$ni = \Phi (\mathcal{R}_c + \mathcal{R}_g) \quad (14.3)$$

Usually, $\mathcal{R}_c \ll \mathcal{R}_g$, and hence Eq. (14.3) can be approximated as

$$ni \approx \Phi \mathcal{R}_g \quad (14.4)$$

The air gap dominates the inductor properties. Four design constraints now can be identified.

14.1.1 Maximum Flux Density

Given a peak winding current I_{max} , it is desired to operate the core flux density at a maximum value B_{max} . The value of B_{max} is chosen to be less than the worst-case saturation flux density B_{sat} of the core material.

Substitution of $\Phi = BA_c$ into Eq. (14.4) leads to

$$ni = BA_c \mathcal{R}_g \quad (14.5)$$

Upon letting $I = I_{max}$ and $B = B_{max}$, we obtain

$$nI_{max} = B_{max} A_c \mathcal{R}_g = B_{max} \frac{\ell_g}{\mu_0} \quad (14.6)$$

This is the first design constraint. The turns ratio n and the air gap length ℓ_g are unknowns.

14.1.2 Inductance

The given inductance value L must be obtained. The inductance is equal to

$$L = \frac{n^2}{\mathcal{R}_g} = \frac{\mu_0 A_c n^2}{\ell_g} \quad (14.7)$$

This is the second design constraint. The turns ratio n , core area A_c , and gap length ℓ_g are unknown.

14.1.3 Winding Area

As illustrated in Fig. 14.5, the winding must fit through the window, i.e., the hole in the center of the core. The cross-sectional area of the conductor, or bare area, is A_w . If the winding has n turns, then the area of copper conductor in the window is

$$nA_w \quad (14.8)$$

If the core has window area W_A , then we can express the area available for the winding conductors as

$$K_u W_A \quad (14.9)$$

where K_u is the *window utilization factor*, or *fill factor*. Hence, the third design constraint can be expressed as

$$K_u W_A \geq nA_w \quad (14.10)$$

The fill factor K_u is the fraction of the core window area that is filled with copper. K_u must lie between zero and one. As discussed in [1], there are several mechanism that cause K_u to be less than unity. Round wire does not pack perfectly; this reduces K_u by a factor of 0.7 to 0.55, depending on the winding technique. The wire has insulation; the ratio of wire conductor area to total wire area varies from approximately 0.95 to 0.65, depending on the wire size and type of insulation. The bobbin uses some of the window area. Insulation may be required between windings and/or winding layers. Typical values of K_u for cores with winding bobbins are: 0.5 for a simple low-voltage inductor, 0.25 to 0.3 for an off-line transformer, 0.05 to 0.2 for a high-voltage transformer supplying several kV, and 0.65 for a low-voltage foil transformer or inductor.

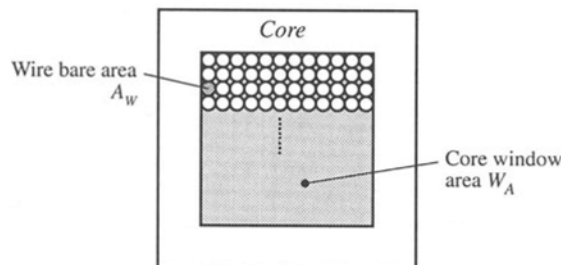


Fig. 14.5 The winding must fit in the core window area.

14.1.4 Winding Resistance

The resistance of the winding is

$$R = \rho \frac{\ell_b}{A_w} \quad (14.11)$$

where ρ is the resistivity of the conductor material, ℓ_b is the length of the wire, and A_w is the wire bare area. The resistivity of copper at room temperature is $1.724 \cdot 10^{-6} \Omega\text{-cm}$. The length of the wire comprising an n -turn winding can be expressed as

$$\ell_b = n(MLT) \quad (14.12)$$

where (MLT) is the mean-length-per-turn of the winding. The mean-length-per-turn is a function of the core geometry. Substitution of Eq. (14.12) into (14.11) leads to

$$R = \rho \frac{n(MLT)}{A_w} \quad (14.13)$$

This is the fourth constraint.

14.1.5 The Core Geometrical Constant K_g

The four constraints, Eqs. (14.6), (14.7), (14.10), and (14.13), involve the quantities A_c , W_A , and MLT , which are functions of the core geometry, the quantities I_{max} , B_{max} , μ_0 , L , K_u , R , and ρ , which are given specifications or other known quantities, and n , ℓ_g , and A_w , which are unknowns. Elimination of the unknowns n , ℓ_g , and A_w leads to the following equation:

$$\frac{A_c^2 W_A}{(MLT)} \geq \frac{\rho L^2 I_{max}^2}{B_{max}^2 R K_u} \quad (14.14)$$

The quantities on the right side of this equation are specifications or other known quantities. The left side of the equation is a function of the core geometry alone. It is necessary to choose a core whose geometry satisfies Eq. (14.14).

The quantity

$$K_g = \frac{A_c^2 W_A}{(MLT)} \quad (14.15)$$

is called the core geometrical constant. It is a figure-of-merit that describes the effective electrical size of magnetic cores, in applications where copper loss and maximum flux density are specified. Tables are included in Appendix D that list the values of K_g for several standard families of ferrite cores. K_g has dimensions of length to the fifth power.

Equation (14.14) reveals how the specifications affect the core size. Increasing the inductance or peak current requires an increase in core size. Increasing the peak flux density allows a decrease in core size, and hence it is advantageous to use a core material that exhibits a high saturation flux density. Allowing a larger winding resistance R , and hence larger copper loss, leads to a smaller core. Of course,

the increased copper loss and smaller core size will lead to a higher temperature rise, which may be unacceptable. The fill factor K_u also influences the core size.

Equation (14.15) reveals how core geometry affects the core capabilities. An inductor capable of meeting increased electrical requirements can be obtained by increasing either the core area A_c , or the window area W_A . Increase of the core area requires additional iron core material. Increase of the window area implies that additional copper winding material is employed. We can trade iron for copper, or vice versa, by changing the core geometry in a way that maintains the K_g of Eq. (14.15).

14.2 A STEP-BY-STEP PROCEDURE

The procedure developed in Section 14.1 is summarized below. This simple filter inductor design procedure should be regarded as a first-pass approach. Numerous issues have been neglected, including detailed insulation requirements, conductor eddy current losses, temperature rise, roundoff of number of turns, etc.

The following quantities are specified, using the units noted:

Wire resistivity	ρ	(Ω -cm)
Peak winding current	I_{max}	(A)
Inductance	L	(H)
Winding resistance	R	(Ω)
Winding fill factor	K_u	
Maximum operating flux density	B_{max}	(T)

The core dimensions are expressed in cm:

Core cross-sectional area	A_c	(cm ²)
Core window area	W_A	(cm ²)
Mean length per turn	MLT	(cm)

The use of centimeters rather than meters requires that appropriate factors be added to the design equations.

1. Determine core size

$$K_g \geq \frac{\rho L^2 I_{max}^2}{B_{max}^2 R K_u} 10^8 \quad (\text{cm}^5) \quad (14.16)$$

Choose a core which is large enough to satisfy this inequality. Note the values of A_c , W_A , and MLT for this core. The resistivity ρ of copper wire is $1.724 \cdot 10^{-6} \Omega$ -cm at room temperature, and $2.3 \cdot 10^{-6} \Omega$ -cm at 100°C.

2. Determine air gap length

$$\ell_g = \frac{\mu_0 L I_{max}^2}{B_{max}^2 A_c} 10^4 \quad (\text{m}) \quad (14.17)$$

with A_c expressed in cm². $\mu_0 = 4\pi \cdot 10^{-7}$ H/m. The air gap length is given in meters. The value expressed in Eq. (14.17) is approximate, and neglects fringing flux and other nonidealities.

Core manufacturers sell gapped cores. Rather than specifying the air gap length, the equivalent quantity A_L is used. A_L is equal to the inductance, in mH, obtained with a winding of 1000 turns. When A_L is specified, it is the core manufacturer's responsibility to obtain the correct gap length. Equation (14.17) can be modified to yield the required A_L , as follows:

$$A_L = \frac{10B_{max}^2 A_c^2}{L I_{max}^2} \quad (\text{mH/1000 turns}) \quad (14.18)$$

where A_c is given in cm^2 , L is given in Henries, and B_{max} is given in Tesla.

3. *Determine number of turns*

$$n = \frac{L I_{max}}{B_{max} A_c} 10^4 \quad (14.19)$$

4. *Evaluate wire size*

$$A_w \leq \frac{K_u W_A}{n} \quad (\text{cm}^2) \quad (14.20)$$

Select wire with bare copper area less than or equal to this value. An American Wire Gauge table is included in Appendix D.

As a check, the winding resistance can be computed:

$$R = \frac{\rho n (MLT)}{A_w} \quad (\Omega) \quad (14.21)$$

14.3 MULTIPLE-WINDING MAGNETICS DESIGN VIA THE K_g METHOD

The K_g method can be extended to the case of multiple-winding magnetics, such as the transformers and coupled inductors described in Sections 13.5.3 to 13.5.5. The desired turns ratios, as well as the desired winding voltage and current waveforms, are specified. In the case of a coupled inductor or flyback transformer, the magnetizing inductance is also specified. It is desired to select a core size, number of turns for each winding, and wire sizes. It is also assumed that the maximum flux density B_{max} is given.

With the K_g method, a desired copper loss is attained. In the multiple-winding case, each winding contributes some copper loss, and it is necessary to allocate the available window area among the various windings. In Section 14.3.1 below, it is found that total copper loss is minimized if the window area is divided between the windings according to their apparent powers. This result is employed in the following sections, in which an optimized K_g method for coupled inductor design is developed.

14.3.1 Window Area Allocation

The first issue to settle in design of a multiple-winding magnetic device is the allocation of the window area A_w among the various windings. It is desired to design a device having k windings with turns ratios $n_1 : n_2 : \dots : n_k$. These windings must conduct rms currents $I_1, I_2, \dots, I_k$ respectively. It should be noted that the windings are effectively in parallel: the winding voltages are ideally related by the turns ratios

D.2 EE CORE DATA

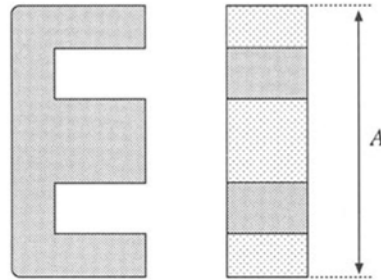


Fig. D.2

Core type	Geometrical constant	Geometrical constant	Cross-sectional area	Bobbin winding area	Mean length per turn	Magnetic path length	Core weight
(A)	K_g	K_{gfe}	A_c	W_A	MLT	ℓ_m	(g)
(mm)	(cm ⁵)	(cm ⁵)	(cm ²)	(cm ²)	(cm)	(cm)	
EE12	$0.731 \cdot 10^{-3}$	$0.458 \cdot 10^{-3}$	0.14	0.085	2.28	2.7	2.34
EE16	$2.02 \cdot 10^{-3}$	$0.842 \cdot 10^{-3}$	0.19	0.190	3.40	3.45	3.29
EE19	$4.07 \cdot 10^{-3}$	$1.3 \cdot 10^{-3}$	0.23	0.284	3.69	3.94	4.83
EE22	$8.26 \cdot 10^{-3}$	$1.8 \cdot 10^{-3}$	0.41	0.196	3.99	3.96	8.81
EE30	$85.7 \cdot 10^{-3}$	$6.7 \cdot 10^{-3}$	1.09	0.476	6.60	5.77	32.4
EE40	0.209	$11.8 \cdot 10^{-3}$	1.27	1.10	8.50	7.70	50.3
EE50	0.909	$28.4 \cdot 10^{-3}$	2.26	1.78	10.0	9.58	116
EE60	1.38	$36.4 \cdot 10^{-3}$	2.47	2.89	12.8	11.0	135
EE70/68/19	5.06	$75.9 \cdot 10^{-3}$	3.24	6.75	14.0	18.0	280

D.6 AMERICAN WIRE GAUGE DATA

AWG#	Bare area, 10^{-3} cm^2	Resistance, $10^{-6} \Omega/\text{cm}$	Diameter, cm
0000	1072.3	1.608	1.168
000	850.3	2.027	1.040
00	674.2	2.557	0.927
0	534.8	3.224	0.825
1	424.1	4.065	0.735
2	336.3	5.128	0.654
3	266.7	6.463	0.583
4	211.5	8.153	0.519
5	167.7	10.28	0.462
6	133.0	13.0	0.411
7	105.5	16.3	0.366
8	83.67	20.6	0.326
9	66.32	26.0	0.291
10	52.41	32.9	0.267
11	41.60	41.37	0.238
12	33.08	52.09	0.213
13	26.26	69.64	0.190
14	20.02	82.80	0.171
15	16.51	104.3	0.153
16	13.07	131.8	0.137
17	10.39	165.8	0.122
18	8.228	209.5	0.109
19	6.531	263.9	0.0948
20	5.188	332.3	0.0874
21	4.116	418.9	0.0785
22	3.243	531.4	0.0701
23	2.508	666.0	0.0632
24	2.047	842.1	0.0566
25	1.623	1062.0	0.0505
26	1.280	1345.0	0.0452
27	1.021	1687.6	0.0409
28	0.8046	2142.7	0.0366
29	0.6470	2664.3	0.0330

Continued

AWG#	Bare area, 10^{-3} cm^2	Resistance, $10^{-6} \Omega/\text{cm}$	Diameter, cm
30	0.5067	3402.2	0.0294
31	0.4013	4294.6	0.0267
32	0.3242	5314.9	0.0241
33	0.2554	6748.6	0.0236
34	0.2011	8572.8	0.0191
35	0.1589	10849	0.0170
36	0.1266	13608	0.0152
37	0.1026	16801	0.0140
38	0.08107	21266	0.0124
39	0.06207	27775	0.0109
40	0.04869	35400	0.0096
41	0.03972	43405	0.00863
42	0.03166	54429	0.00762
43	0.02452	70308	0.00685
44	0.0202	85072	0.00635

REFERENCES

- [1] C. W. T. MCLYMAN, *Transformer and Inductor Design Handbook*, Second edition, New York: Marcel Dekker, 1988.
- [2] *Ferrite Materials and Components Catalog*, Philips Components.